UNITED STATES SPACE FORCE
SPACE OPERATIONS COMMAND

July 24, 2025

FIRST QUARTER LAUNCH RECORD:

There were 61 launches in the first quarter of 2024, surpassing the previous record of 48 in the first quarter of 2023.

SPACE DOMAIN AWARENESS OPERATIONS:

- 9% increase since 31 Dec 2021 – 41,400
- 41% increase since 2020 – 32,000
- 78% increase since 2019 – 25,300
- 91% increase since 2016 – 23,500
- 173% increase since 2011 – 16,499
- 10,000 active spacecraft

A CONGESTED SPACE DOMAIN – 2008 to 2024

TOTAL TRACKED OBJECTS

45,100 • ↑196%

ACTIVE SATELLITES ON ORBIT

10,200 • ↑317%

WORLDWIDE LAUNCHES

177 in 2023 • ↑203% per year
105 in 2024 so far

SATELLITES LAUNCHED

2,621 in 2023 • ↑2544% per year
1022 in 2024 so far

DEBRIS CONJUNCTIONS

1,330 in 2023 • ↑1108% per year
533 in 2024 so far

Data Mine of the Rockies Project 2025-26:

Processing Uncorrelated Tracks (UCTs) in the Common Task Framework: Benchmark Dataset Generation and Evaluation Criteria

©-Josep M. Drudis, 2017

SPACEOPERATIONSCOMMAND

SPACE MISSION DELTAS AND SPACE BASE DELTAS

SPACE DELTA 2 • SDA
Space Domain Awareness | Peterson SFB

Integrates ISR, space observation and environmental monitoring to enable space battle management and support ground operations

SPACE DELTA 4 • MW & MT
Missile Warning & Missile Tracking | Buckley SFB

Provides strategic and theater missile warning to the United States and our International Partners

SPACE DELTA 7 • ISR
Intelligence, Surveillance & Reconnaissance | Peterson SFB

Conducts ISR ops for and from the Space domain to characterize threats to and identify vulnerabilities within Space Delta tactical units, Intelligence Community partners, and Joint warfighters worldwide

SPACE DELTA 9 • OW
Orbital Warfare | Schriever SFB

Conducts protect and defend operations from space and provides response options to deter and defeat adversary threats in space

SPACE BASE DELTA 1
Colorado Springs CO

Peterson SFB, Schriever SFB, Pituffik SB, Cheyenne Mountain SFS, Kaena Point SFS, New Boston SFS and 22 USSF mission locations worldwide

SPACE OPERATIONS COMMAND

SpOC
Peterson SFB

SPACE DELTA 3 • SEW
Space Electromagnetic Warfare | Peterson SFB

Operates electromagnetic attack, protection, and support capabilities to protect and defend the space domain

SPACE DELTA 6 • CYBER OPS
Cyberspace Operations | Schriever SFB

Executes cyber operations to protect space operations, networks, and communications, and operates the Satellite Control Network

SPACE DELTA 8 • SATCOM
Satellite Communications | Schriever SFB

Delivers assured, effective, and resilient Satellite Communications for our Nation to connect and project space capabilities worldwide

PNT DELTA (P) • NAVWAR
Navigational Warfare | Schriever SFB

Provide, operate, and sustain, high-integrity Positioning, Navigation, & Timing capabilities to protect our Nation's interests and assure an unparalleled global utility

SPACE BASE DELTA 2
Aurora CO

Buckley SFB, Cape Cod SFS, Cavalier SFS, Clear SFS and 10 USSF mission locations worldwide

SPACE FUELS AMERICANS' WAY OF LIFE... AND IS VITAL TO OUR NATIONAL DEFENSE

SPACE IS CRITICAL TO OUR WAY OF LIFE:

- It is estimated the global space industry could generate more than $1 trillion or more in revenue by 2040, up from current $350 billion.
- The USSF operates the Global Positioning System (GPS), one of several satellite constellations, as a global utility used by over a billion users and available to everyone, everywhere on Earth.

SPACE IS CRITICAL TO OUR MILITARY:

- The GPS constellation enables 716 different weapon systems fielded by the DoD, and approximately six billion civil users globally that leverage GEOINT products.
- The United States military is designed to have unfettered access to space capabilities, an assumption that is challenged daily by adversaries around the world.

USSF's MISSION IS TO "SECURE OUR NATION'S INTERESTS IN, FROM, AND TO SPACE."

ASSURED SPACE ACCESS
"ACCESS THE DOMAIN"

- Deploy and sustain equipment in space
- Missions: Launch, Range Control, Satellite Control Network

SPACE SUPERIORITY
"CONTROL THE DOMAIN"

- Defend against space and counterspace threats
- Missions: Orbital Warfare, Electromagnetic Warfare, Space Battle Management

GLOBAL MISSION OPERATIONS
"EXPLOIT THE DOMAIN"

- Globally integrate joint functions across all domains
- Missions: Missile Warning, Satellite Communications, Positioning, Navigation, and Timing (PNT)

Orbital Warfare

Space Battle Management

Electromagnetic Warfare

Engineering & Acquisition

Space Access & Sustainment

Military Intelligence

Cyber Operations

Project Background: 1

The Space Operations Command (SpOC), in partnership with The SDA TAP Lab, is undertaking the development of a fully automated data annotation pipeline to facilitate the training and evaluation of machine learning algorithms for space domain awareness (SDA) tasks. This follows many industry best practices which are summarized in "50 Years of Data Science" as the Common Task Framework (Donoho, 2017).

Currently, the open-source development community focused on the space domain is very small. There are no widely accepted ontologies for common tasks, labels, benchmarks, or performance metrics. As a result, we cannot say with any confidence what the state-of-the-art performance is for any algorithm employed in our enterprise. This is a challenge.

The overall goal of the project is to accelerate the delivery of space battle management software to operational units by automating Common Tasks throughout the Kill Chain. Particular common tasks of high priority include Uncorrelated Track (UCT) Processing, State Estimation, and Object Identification; a more comprehensive list of tasks can be found at sdataplab.org. The Common Task Framework, as defined by Donoho's 2017 paper 50 Years of Data Science, provides a standardized method to evaluate and compare the performance of such automation algorithms.

©-Josep M. Drudis, 2017

Project Background: 2

A wealth of information related to these tasks exists on the Unified Data Library (UDL) and other online data sources, including observational data for both known objects and UCTs, current TLE and state vectors for tracked objects, launch events, maneuver events, etc. In order to use this data, software tools must be developed to streamline or automate the creation of benchmark datasets used for training and evaluation in the Common Task Framework. These software tools must be able to pull event data from available data sources, label that data according to predefined classifications (i.e. launch events, maneuver events, proximity events, etc.), parse and/or extract relevant measurement data depending on the classification, clean and store the labelled data in a centrally available database. As an iterative process over time, as subject matter experts refine data label definitions, higher fidelity data will become available. From this database, multiple benchmark datasets can be generated to either train or test various algorithms developed to automate common task workflows.

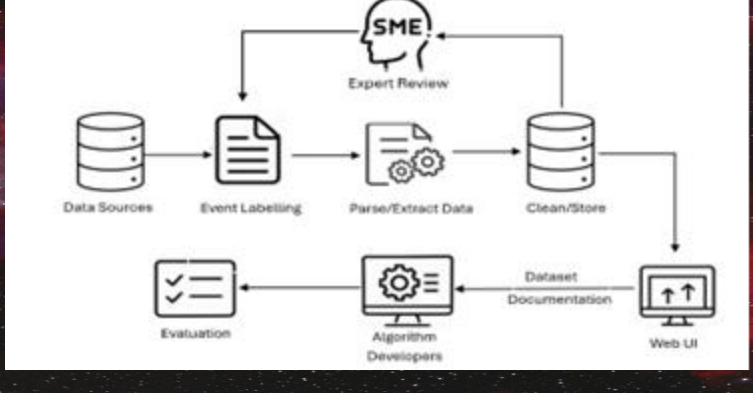
The process of labelling and storing data (the goals of the sister project of the SDA TAP Lab), as well as benchmark dataset generation and evaluation criteria (the goal of this project), must be well documented and repeatable so that it can be generalized and applied to other common tasks as they are identified. Additionally, documentation detailing the backend functionality of dataset generation and evaluation criteria must be made available to algorithm developers to enable a clear understanding of how algorithms can be improved.

Robert M. Drudis, 2017

The end result of the Common Task Framework defined above is a Web-hosted User Interface that algorithm developers can interact with to gen

Project Goal

Develop benchmark dataset generation and evaluation criteria for processing Uncorrelated Tracks (UCTs) in the Common Task Framework. The end result of the Common Task Framework is a Web-hosted User Interface that algorithm developers can interact with to generate and download benchmark datasets related to the common task being solved in order to train their algorithms, as well as upload the results of the solution presented to be objectively evaluated and compared to other solutions presented to the same common task.



©-Josep M. Drudis, 2017